

# VAANI GOENKA

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## EDUCATION

**Ashoka University, 2024-2028:** B. Sc. Computer Science and Mathematics. **Received a 100% scholarship for merit in Computer Science.**

**GPA: 3.95 / 4.00.** Relevant coursework: Cryptography, Data Structures & Algorithms, Design & Analysis of Algorithms, Theory of Computation, Symbolic Logic and Applications, Multivariate Calculus

Modern High School for Girls, 2010 - 2024: **ICSE:** 97% (Mar 2022) | **ISC:** 98.25% (Mar 2024), School Rank 4 | **General Proficiency award** (Highest Academic Award), All-Rounder Achievement

**SAT: 1580/1600.** Advanced Placement (AP) Test: Computer Science A: 5, Calculus AB: 5, English Language and Composition: 5.

## PUBLICATIONS

- Goenka, V., Thakkar, A. (2026). BOOP: Write Right Code. In: Prasad, P., Raman, A., Shah, B. (eds) Computing Education Research . COMPUTE 2025. Communications in Computer and Information Science, vol 2739. Springer, Cham. [https://doi.org/10.1007/978-3-032-14583-3\\_7](https://doi.org/10.1007/978-3-032-14583-3_7)
- Vaani Goenka and Aalok Thakkar. 2026. A Correctness-First Introduction to Computer Science. In Proceedings of the 31st ACM Conference on Innovation and Technology in Computer Science Education V. 1 (ITiCSE 2026). Association for Computing Machinery, New York, NY, USA, 345–351. <https://doi.org/10.1145/3803400.3809367>

## EXPERIENCE

- Research Assistant, School of Technology and Computer Science, Tata Institute of Fundamental Research** May 2025 - Current
- Working under Prof. Shibashis Guha on Model-Checking in Markov Decision Processes.
- Research Assistant, MPhasis Lab, "AI At Ashoka"** May 2025 - Current
- Co-authored "BOOP: Write Right Code", a research paper on CS Pedagogy and Program Correctness, accepted at ACM COMPUTE 2025.
  - Developed a Visual Studio Extension, leveraging programming language parsers, automated theorem provers and on-ground data. Moreover, developed instructor and student interfaces for optimal adoption.
  - Currently working on automated grading using automated tools for verifying code and related formal methods.
- Teaching Assistant, Ashoka University** May 2025 - Dec 2025
- Was a teaching Assistant in the Lodha Genius Program for "Introduction to Modern Cryptography", "Linear Algebra", "Graph Theory & Recurrence Relations" and "Problem Solving Group" (Mathematics Olympiad Style Questions).
  - Led weekly problem-solving sessions and prepared material for the above courses.
- Yugen Space, Software Engineering Intern - LLMs and Ai** December 2024 - February 2025
- Developed pipelines for RAG systems, database integration, and frontend workflows for an AI-integrated Geospatial Data Platform.
  - Built geo-spatial data visualization with LLM-enabled interaction, enabling multimodal (limited) queries.
- Ashoka University Ministry of Technology, Deputy Minister** August 2024 - Ongoing
- Designed, developed and deployed an end-to-end solution for automating the Mailroom package system at Ashoka University – currently in use by 4000 students, 10 staff members, helping them reduce time spent per package by 70%; **received a 'Student Life Service' Award**. Developed a gamified notification system; responsible for ongoing server management and maintenance.
  - Led a team of 10 developers to build micro and macro digital infrastructure for the student body and maintain the systems in place, working closely with the IT and Administration teams.
  - Revamped legacy platforms using a full-stack React/Next.js/Strapi framework, improving design and UI while incorporating iterative user feedback.

## AWARDS & HONORS

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|-------------------------------------------------------------------------------------|------|
| RSR Grant for Women in STEM Research                                                | 2026 |
| Student Life Service Award (Given to the Ministry of Technology, Ashoka University) | 2025 |
| Runner-Up   Apart Research Sprint (Artificial Intelligence and AI Safety)           | 2025 |

## TEACHING

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|--------------------------------|------------------------------------------------------------------------------|
| Discrete Mathematics           | (Prof. Gaurav Bhatnagar: Monsoon 2025, Prof. Partha Pratim Das: Spring 2026) |
| Data Structures and Algorithms | (Prof. Aalok Thakkar: Monsoon 2026)                                          |